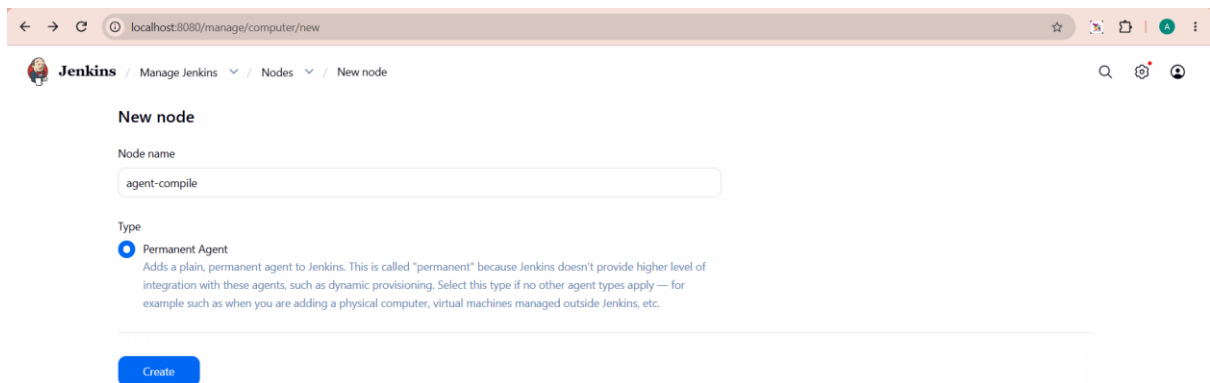


Project 4: Architecting Jenkins Pipeline for Scale-Use Jenkins to set up a distributed pipeline that will compile and test a Maven project on two different slave nodes respectively

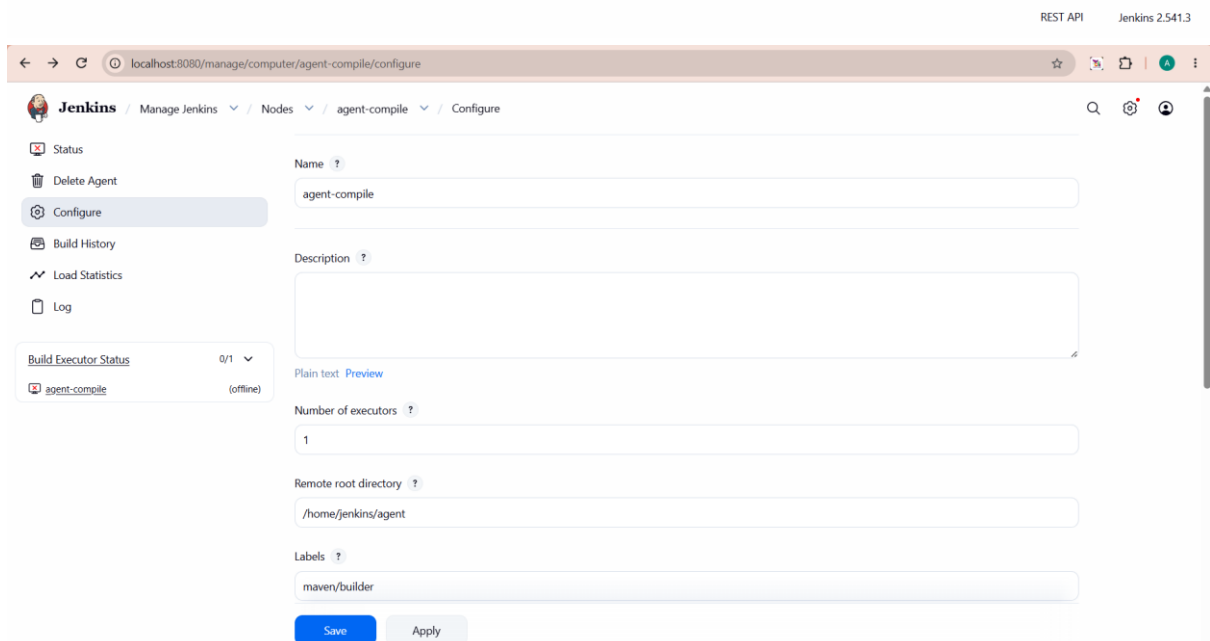
Name: Aarya Balwadkar

PRN: 23070122005

1. Create Agent Compile



The screenshot shows the Jenkins 'New node' configuration page. The browser address bar is 'localhost:8080/manage/computer/new'. The page title is 'New node'. The 'Node name' field contains 'agent-compile'. The 'Type' section has 'Permanent Agent' selected with a radio button. Below it, a description states: 'Adds a plain, permanent agent to Jenkins. This is called "permanent" because Jenkins doesn't provide higher level of integration with these agents, such as dynamic provisioning. Select this type if no other agent types apply — for example such as when you are adding a physical computer, virtual machines managed outside Jenkins, etc.' At the bottom, there is a blue 'Create' button.



The screenshot shows the Jenkins 'agent-compile/configure' page. The browser address bar is 'localhost:8080/manage/computer/agent-compile/configure'. The page title is 'Configure'. On the left, there is a sidebar with links: 'Status', 'Delete Agent', 'Configure' (highlighted), 'Build History', 'Load Statistics', and 'Log'. Below the sidebar, there is a 'Build Executor Status' section showing '0/1' and a dropdown menu with 'agent-compile (offline)'. The main configuration area has the following fields: 'Name' (agent-compile), 'Description' (empty), 'Number of executors' (1), 'Remote root directory' (/home/jenkins/agent), and 'Labels' (maven/builder). At the bottom, there are 'Save' and 'Apply' buttons. The top right corner shows 'REST API' and 'Jenkins 2.541.3'.

localhost:8080/manage/computer/agent-compile/configure

Jenkins

Manage Jenkins

Nodes

agent-compile

Configure

Labels ?

maven/builder

Usage ?

Use this node as much as possible

Launch method ?

Launch agent by connecting it to the controller

Availability ?

Keep this agent online as much as possible

Node Properties

☐ Disable deferred wipeout on this node ?

☐ Disk Space Monitoring Thresholds

☐ Environment variables

☐ Tool Locations

Save

Apply

```
PS C:\Users\Aarya> docker network connect jenkins-net myjenkins
PS C:\Users\Aarya> docker run -d --name agent-compile --network jenkins-net jenkins/inbound-agent:latest -url http://myjenkins:8080 3d63d14d96924530659850dd
41990e68ef2fb6359b2b276c637e7f52f47bd850 agent-compile
Unable to find image 'jenkins/inbound-agent:latest' locally
latest: Pulling from jenkins/inbound-agent
431257b45158: Pull complete
05daae01d1b4: Pull complete
56c4fac4b0ad: Pull complete
308b9187121e: Pull complete
c6cc732f5fb2: Pull complete
26c307b5e35a: Pull complete
f16ede14786f: Pull complete
4f4fb700ef54: Pull complete
1939c9227a43: Pull complete
8b9d72553e6f: Pull complete
d7996413ef54: Pull complete
47ebf3e1f03e: Download complete
Digest: sha256:6d8eabd48bb3926293de0b337117f7ad8bb2c0ba1a425b3e34e3d3983b5e1f9
Status: Downloaded newer image for jenkins/inbound-agent:latest
dfe6d194a7e70d02655cef2eb19ce181631268a415d38bed9e4f6653f44f075
PS C:\Users\Aarya>
```

localhost:8080/computer/

Jenkins

Nodes

+ New Node

Configure Monitors

Nodes

Add, remove, control and monitor the various nodes that Jenkins runs jobs on.

S	Name	Architecture	Clock Difference	Free Disk Space	Free Swap Space	Free Temp Space	Response Time
	agent-compile	Linux (amd64)	In sync	947.36 GiB	1023.74 MiB	947.36 GiB	94ms
	Built-In Node	Linux (amd64)	In sync	947.36 GiB	1023.74 MiB	947.36 GiB	0ms
last checked		5.4 sec	5.4 sec	5.3 sec	5.3 sec	5.3 sec	5.4 sec

Icon:

S

M

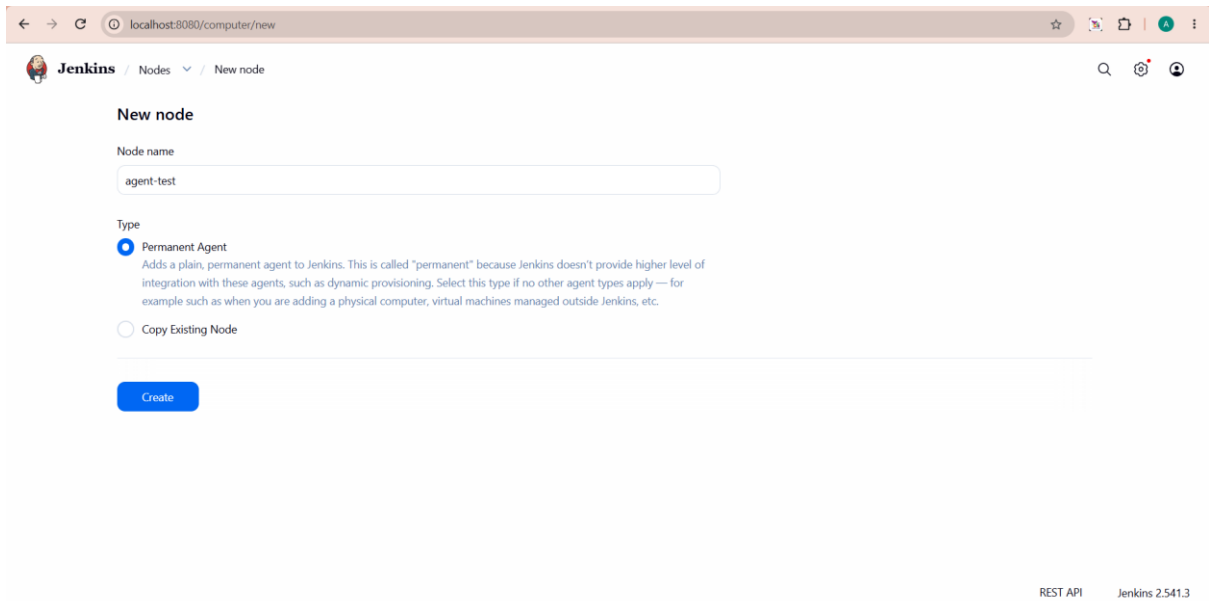
L

Legend

REST API

Jenkins 2.541.3

2. Create Agent Test



This screenshot shows the 'New node' page in Jenkins. The browser address bar indicates the URL is localhost:8080/computer/new. The page title is 'New node'. Under 'Node name', the text 'agent-test' is entered. Under 'Type', the 'Permanent Agent' radio button is selected. A descriptive text for 'Permanent Agent' is provided: 'Adds a plain, permanent agent to Jenkins. This is called "permanent" because Jenkins doesn't provide higher level of integration with these agents, such as dynamic provisioning. Select this type if no other agent types apply — for example such as when you are adding a physical computer, virtual machines managed outside Jenkins, etc.' The 'Copy Existing Node' radio button is unselected. A blue 'Create' button is at the bottom. The bottom right corner shows 'REST API' and 'Jenkins 2.541.3'.

localhost:8080/computer/new

Jenkins / Nodes / New node

New node

Node name

agent-test

Type

☒ Permanent Agent

Adds a plain, permanent agent to Jenkins. This is called "permanent" because Jenkins doesn't provide higher level of integration with these agents, such as dynamic provisioning. Select this type if no other agent types apply — for example such as when you are adding a physical computer, virtual machines managed outside Jenkins, etc.

☐ Copy Existing Node

Create

REST API Jenkins 2.541.3



This screenshot shows the 'createitem' page in Jenkins. The browser address bar indicates the URL is localhost:8080/computer/createitem. The page title is 'Jenkins / Nodes'. The 'Name' field contains 'agent-test'. The 'Description' field is empty. Below the description is a 'Plain text Preview' link. The 'Number of executors' field contains '1'. The 'Remote root directory' field contains '/home/jenkins/agent'. The 'Labels' field contains 'maven-tester'. A blue 'Save' button is at the bottom.

localhost:8080/computer/createitem

Jenkins / Nodes

Name ?

agent-test

Description ?

Plain text Preview

Number of executors ?

1

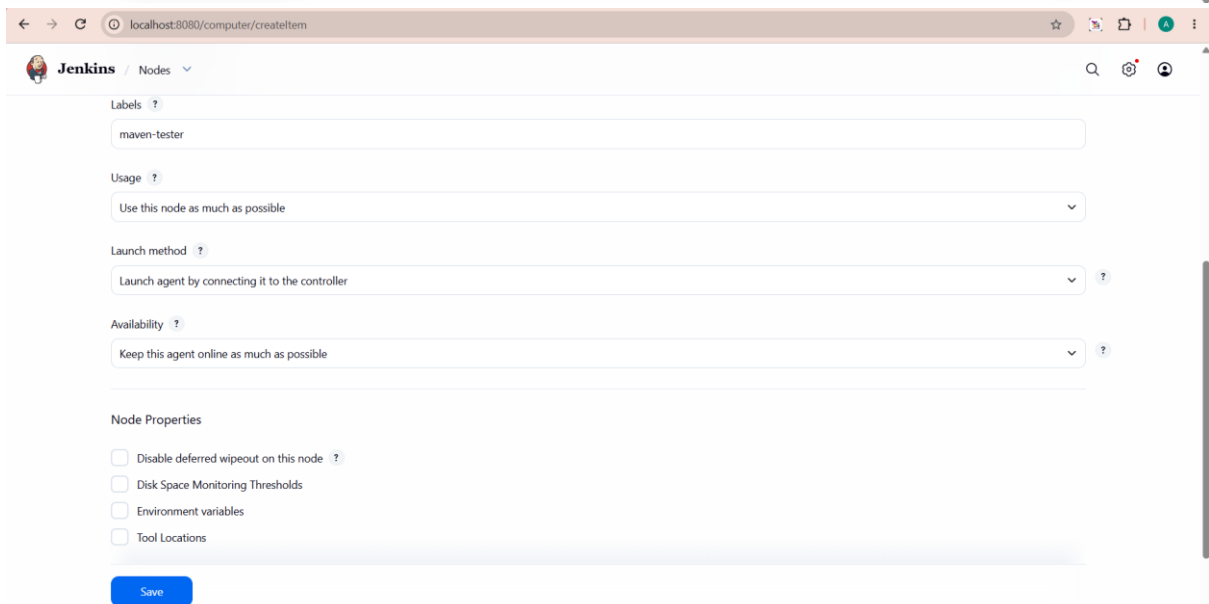
Remote root directory ?

/home/jenkins/agent

Labels ?

maven-tester

Save



This screenshot shows the 'createitem' page in Jenkins, continuing from the previous one. The 'Labels' field contains 'maven-tester'. The 'Usage' dropdown menu is set to 'Use this node as much as possible'. The 'Launch method' dropdown menu is set to 'Launch agent by connecting it to the controller'. The 'Availability' dropdown menu is set to 'Keep this agent online as much as possible'. Under 'Node Properties', there are four unchecked checkboxes: 'Disable deferred wipeout on this node', 'Disk Space Monitoring Thresholds', 'Environment variables', and 'Tool Locations'. A blue 'Save' button is at the bottom.

localhost:8080/computer/createitem

Jenkins / Nodes

Labels ?

maven-tester

Usage ?

Use this node as much as possible

Launch method ?

Launch agent by connecting it to the controller

Availability ?

Keep this agent online as much as possible

Node Properties

☐ Disable deferred wipeout on this node ?

☐ Disk Space Monitoring Thresholds

☐ Environment variables

☐ Tool Locations

Save

```
PS C:\Users\Aarya> docker run -d --name agent-test --network jenkins-net jenkins/inbound-agent:latest -url http://myjenkins:8080 e58a72b2e0ae6f2a3b70abdac4bfe4f1d9641c09634f16e95dfc07ba8d5a84 agent-test
03d21140c6efd2381f80a119e6879b3561d2efa13dddb377fd1498cc97b59a79
PS C:\Users\Aarya> |
```

← → ↻ localhost:8080/computer/

Jenkins / Nodes

Nodes [+ New Node](#) [Configure Monitors](#) [↻](#)

Add, remove, control and monitor the various nodes that Jenkins runs jobs on.

S	Name	Architecture	Clock Difference	Free Disk Space	Free Swap Space	Free Temp Space	Response Time
	agent-compile	Linux (amd64)	In sync	947.34 GiB	1023.24 MiB	947.34 GiB	100ms
	agent-test	Linux (amd64)	In sync	947.34 GiB	1023.24 MiB	947.34 GiB	40ms
	Built-In Node	Linux (amd64)	In sync	947.34 GiB	1023.24 MiB	947.34 GiB	0ms
last checked		16 sec	16 sec	16 sec	16 sec	16 sec	16 sec

Icon: S M **L** Legend

REST API Jenkins 2.541.3

3. Install maven on Jenkins

← → ↻ localhost:8080/manage/configureTools/

Jenkins / Manage Jenkins ▾ / Tools

Tools
Configure tools, their locations and automatic installers.

Maven Configuration

Default settings provider
 ?

Default global settings provider
 ?

JDK installations
[+ Add JDK](#)

Git installations

[Save](#) [Apply](#)

← → ↻ localhost:8080/manage/configureTools/

Jenkins / Manage Jenkins ▾ / Tools

Git

Name

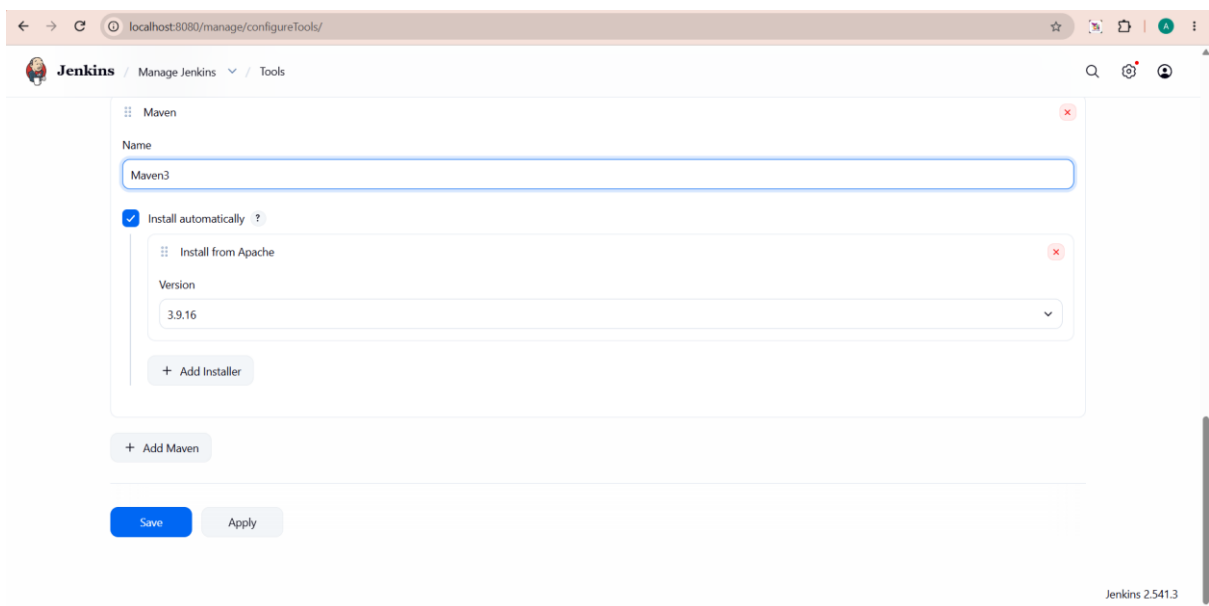
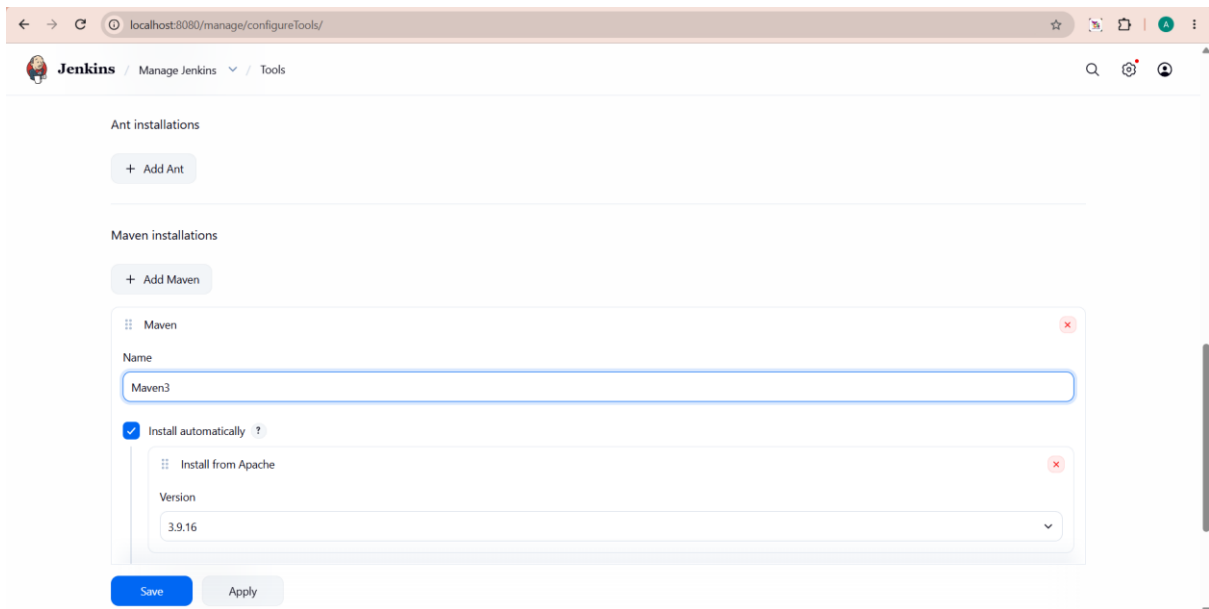
Path to Git executable ?

☐ Install automatically ?

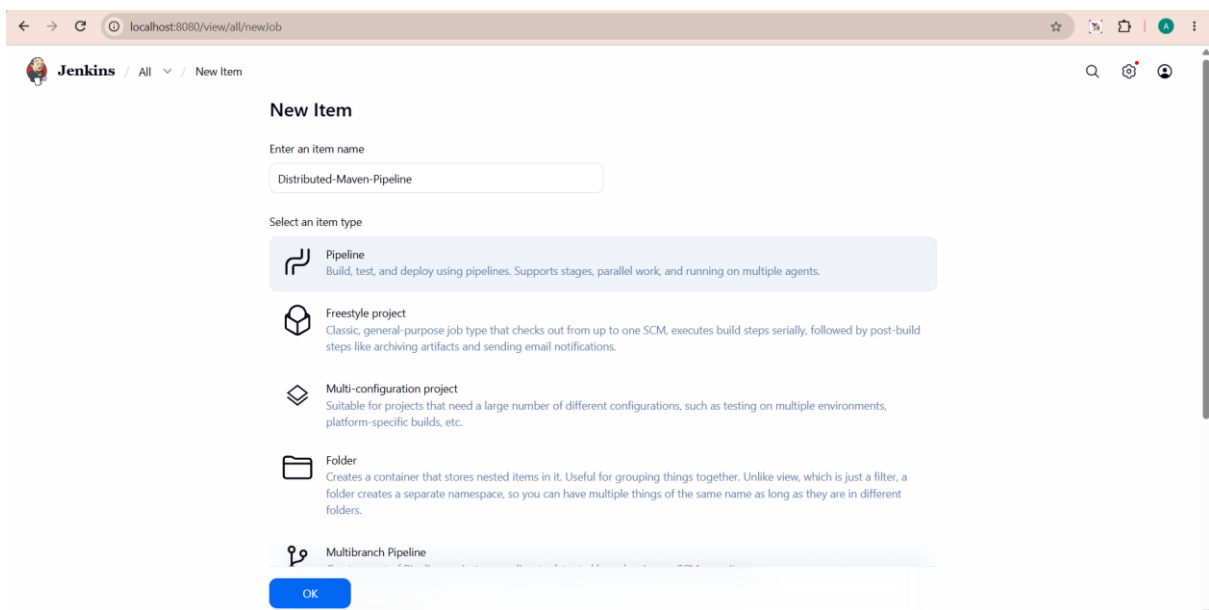
[+ Add Git](#)

Gradle installations
[+ Add Gradle](#)

Ant installations
[Save](#) [Apply](#)



Create new Item



localhost:8080/job/Distributed-Maven-Pipeline/configure

Jenkins / Distributed-Maven-Pipeline / Configure

General Enabled

Description

Distributed Maven Pipeline containing of 2 agents:
1. Maven builder
2. Maven tester

Plain text [Preview](#)

☐ Discard old builds ?

☐ Do not allow concurrent builds

☐ Do not allow the pipeline to resume if the controller restarts

☐ GitHub project

☐ Pipeline speed/durability override ?

☐ Preserve stashes from completed builds ?

☐ This project is parameterised ?

[Save](#) [Apply](#)

localhost:8080/job/Distributed-Maven-Pipeline/configure

Jenkins / Distributed-Maven-Pipeline / Configure

General

Triggers

Set up automated actions that start your build based on specific events, like code changes or scheduled times.

☐ Build after other projects are built ?

☐ Build periodically ?

☐ GitHub hook trigger for GITScm polling ?

☐ Poll SCM ?

☐ Trigger builds remotely (e.g. from scripts) ?

Pipeline

Define your Pipeline using Groovy directly or pull it from source control.

Definition

Pipeline script

Script ?

30 label 'maven-tester' [try sample Pipeline...](#)

[Save](#) [Apply](#)

localhost:8080/job/Distributed-Maven-Pipeline/configure

Jenkins / Distributed-Maven-Pipeline / Configure

General

Triggers

Pipeline

Definition

Pipeline script

Script ?

```
1 pipeline {  
2   // We set the global agent to 'none' so the Master node does ZERO work.  
3   agent none  
4  
5   stages {  
6     stage('compile Code') {  
7       // This forces Jenkins to run this stage ONLY on Node 1  
8       agent {  
9         label 'maven-builder'  
10      }  
11     tools {  
12       maven 'Maven3'  
13     }  
14     steps {  
15       echo 'Checking out the code...'  
16     }  
17   }  
18 }  
19
```

[try sample Pipeline...](#)

☒ Use Groovy Sandbox ?

[Pipeline Syntax](#)

[Save](#) [Apply](#)

localhost:8080/job/Distributed-Maven-Pipeline/configure

Jenkins / Distributed-Maven-Pipeline / Configure

Configure

- General
- Triggers
- Pipeline
- Advanced

Definition

Pipeline script

Script ?

```
16      git 'https://github.com/jglick/simple-maven-project-with-tests.git'
17
18      echo 'compiling the application using JDK 17...'
19      sh 'mvn clean compile'
20
21      // IMPORTANT: Node 1 and Node 2 are different computers.
22      // We must "stash" the compiled files so they can be sent across the network.
23      stash name: 'compiled-code', includes: '**/*'
24  }
25  }
26
27  stage('Run Tests') {
28      // This forces Jenkins to run this stage ONLY on Node 2
29      agent {
30          label 'maven-tester'
```

☒ Use Groovy Sandbox ?

[Pipeline Syntax](#)

Save Apply

localhost:8080/job/Distributed-Maven-Pipeline/configure

Jenkins / Distributed-Maven-Pipeline / Configure

Configure

- General
- Triggers
- Pipeline
- Advanced

Definition

Pipeline script

Script ?

```
30      label 'maven-tester'
31  }
32  tools {
33      maven 'Maven3'
34  }
35  steps {
36      // We unpack the compiled code from Node 1
37      unstash 'compiled-code'
38
39      echo 'Testing the compiled application...'
40      sh 'mvn test'
41  }
42  }
43  }
44  }
```

☒ Use Groovy Sandbox ?

[Pipeline Syntax](#)

Save Apply

localhost:8080/job/Distributed-Maven-Pipeline/

Jenkins / Distributed-Maven-Pipeline

Status

- Changes
- Build Now
- Configure
- Delete Pipeline
- Stages
- Rename
- Pipeline Syntax

Distributed-Maven-Pipeline

Distributed Maven Pipeline containing of 2 agents:
1. Maven builder
2. Maven tester

Permalinks

Builds >

Today

#1 03:44

REST API Jenkins 2.541.3

Outputs

localhost:8080/job/Distributed-Maven-Pipeline/1/console

Jenkins / Distributed-Maven-Pipeline #1

Status

Changes

Console Output

Edit Build Information

Git Build Data

Pipeline Overview

Thread Dump

Pause/resume

Replay

Pipeline Steps

Workspaces

Console

Download Copy View as plain text

```
Started by user Aarya Balwadkar
[Pipeline] Start of Pipeline
[Pipeline] stage
[Pipeline] { (Compile Code)
[Pipeline] node
Still waiting to schedule task
There are no nodes with the label 'maven-builder'
Running on agent-compile in /home/jenkins/agent/workspace/Distributed-Maven-Pipeline
[Pipeline] {
[Pipeline] tool
Unpacking https://repo.maven.apache.org/maven2/org/apache/maven/apache-maven/3.9.16/apache-maven-3.9.16-bin.zip to
/home/jenkins/agent/tools/hudson.tasks.Maven_MavenInstallation/Maven3 on agent-compile
[Pipeline] envVarsForTool
[Pipeline] withEnv
[Pipeline] {
[Pipeline] echo
Checking out the code...
[Pipeline] git
The recommended git tool is: NONE
No credentials specified
Cloning the remote Git repository
Cloning repository https://github.com/jglick/simple-maven-project-with-tests.git
> git init /home/jenkins/agent/workspace/Distributed-Maven-Pipeline # timeout=10
Fetching upstream changes from https://github.com/jglick/simple-maven-project-with-tests.git
> git --version # timeout=10
```

localhost:8080/job/Distributed-Maven-Pipeline/1/console

Jenkins / Distributed-Maven-Pipeline #1

```
> git --version # 'git version 2.47.3'
> git fetch --tags --force --progress -- https://github.com/jglick/simple-maven-project-with-tests.git +refs/heads/*:refs/remotes/origin/* #
timeout=10
> git config remote.origin.url https://github.com/jglick/simple-maven-project-with-tests.git # timeout=10
> git config --add remote.origin.fetch +refs/heads/*:refs/remotes/origin/* # timeout=10
Avoid second fetch
Checking out Revision 9a5e9832814aa17fa7e514e5890fc6cf913d9a77 (refs/remotes/origin/master)
> git rev-parse refs/remotes/origin/master^{commit} # timeout=10
> git config core.sparsecheckout # timeout=10
> git checkout -f 9a5e9832814aa17fa7e514e5890fc6cf913d9a77 # timeout=10
> git branch -a -v --no-abbrev # timeout=10
> git checkout -b master 9a5e9832814aa17fa7e514e5890fc6cf913d9a77 # timeout=10
Commit message: "Creating a main class so there is something for 'javadoc:javadoc' to do"
First time build. Skipping changelog.
[Pipeline] echo
Compiling the application using JDK 17...
[Pipeline] sh
+ mvn clean compile
[INFO] Scanning for projects...
[INFO]
[INFO] -----< test:simple-maven-project-with-tests >-----
[INFO] Building simple-maven-project-with-tests 1.0-SNAPSHOT
[INFO] from pom.xml
[INFO] -----[ jar ]-----
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-clean-plugin/3.2.0/maven-clean-plugin-3.2.0.pom
Progress (1): 780 B
Progress (1): 1.7 kB
Progress (1): 4.2 kB
```

localhost:8080/job/Distributed-Maven-Pipeline/1/console

Jenkins / Distributed-Maven-Pipeline #1

```
-----
T E S T S
-----
Running test.OtherTest
Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.001 sec - in test.OtherTest
Running test.SomeTest
Tests run: 6, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.001 sec - in test.SomeTest

Results :

Tests run: 7, Failures: 0, Errors: 0, Skipped: 0

[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 17.058 s
[INFO] Finished at: 2026-08-09T03:51:27Z
[INFO] -----
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // node
[Pipeline] }
[Pipeline] // stage
[Pipeline] End of Pipeline
Finished: SUCCESS
```